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Graphyne Could Be Better Than Graphene

by Jon Cartwright on 1 March 2012, 2:57 PM | [3 Comments](#)[Email](#)[Print](#) |

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Graphene, a layer of graphite just one atom thick, isn't called a wonder material for nothing. The subject of the 2010 Nobel Prize in physics, it is famed for its superlative mechanical and electronic properties. Yet new computer simulations suggest that the electronic properties of a little-known sister material of graphene—graphyne—may in some ways be better.

The simulations show that graphyne's conduction electrons should travel extremely fast—as they do in graphene—but in only one direction. That property could help researchers design faster transistors and other electronic components that process one-way current, says one of the study's authors, theoretical chemist Andreas Görling of the University of Erlangen-Nuremberg in Germany. "If your material already conducts in one direction, you have a head start," he says.

Graphene gets its properties from its unusual structure, in which carbon atoms are bonded together in a hexagonal pattern like atomic-scale chicken wire. The bonds hover halfway between single and double bonds, making them so strong that it's almost impossible to make defects in the lattice.

Electrically, graphene's structure has also been considered unique. In most materials, conduction electrons—the particles that carry electric current—have an energy that depends on the square of their momentum. Graphene's electronic energy levels, however, stack into shapes called Dirac cones, which allow conduction electrons to travel with an energy that is directly proportional to their momentum. As a result, the electrons travel as though they were massless, the way particles of light do—in other words, very fast.

Graphyne is similar to graphene in that it is also a two-dimensional structure of carbon. Unlike graphene, though, graphyne contains double and triple bonds and its atoms do not always have a hexagonal arrangement. Indeed, there may be a vast number of possible graphynes, each with the double and triple bonds in slightly different arrangements. Theorists have been studying graphynes since the 1980s, but little work has been devoted to their electronic properties, Görling says.

Görling's group has now examined these electronic properties in computer simulations, using a technique called density functional theory. This is standard for mapping the energy levels of different possible forms of the material. The researchers discovered that in one particular graphyne—so-called 6,6,12-graphyne, which has a rectangular lattice arrangement—Dirac cones should still exist but in a distorted, squashed form. As a result, the researchers reported last week in *Physical Review Letters*, the material should conduct electrons in a preferred direction.

Karsten Horn, an experimental materials scientist at the Max Planck Society for the Advancement of Science in Berlin, is surprised that Dirac cones—and their associated electrical conductivity—could exist in materials other than graphene, particularly those that lack hexagonal symmetry. But he thinks computer simulations alone are not enough to prove that the electrical properties will exist in practice. "It may be true, but the test of the cooking is in the eating," he says.

Condensed-matter theorist Mikhail Katsnelson of the Radboud University of Nijmegen in the Netherlands agrees that experiments will tell for sure. Still, he has confidence in the computer simulations. "Density-functional calculations are quite reliable," he says. "For instance, hydrogenated graphene [a type used to make graphene transistors] was first predicted by density-functional calculations and then was observed experimentally."

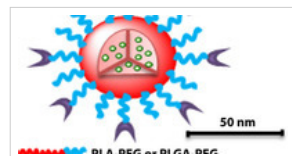
The very first test for graphyne, however, will be to manufacture it. Only one type of graphyne has been claimed to be made in the lab, and it wasn't the 6,6,12-graphyne Görling's group studied. But, says Görling, the promising theoretical results might encourage synthetic chemists to make theirs.

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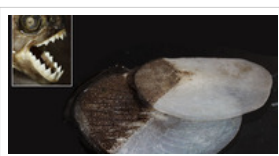
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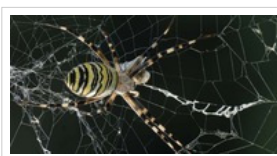
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